

Differences Between Out-of-Hospital Cardiac Arrest in Residential and Public Locations and Implications for Public-Access Defibrillation

Fredrik Folke, MD; Gunnar H. Gislason, MD, PhD; Freddy K. Lippert, MD; Søren L. Nielsen, MD; Peter Weeke, MD; Morten L. Hansen, MD, PhD; Emil L. Fosbøl, MD, PhD; Søren S. Andersen, MD; Søren Rasmussen, MSc, PhD; Tina K. Schramm, MD; Lars Køber, MD, DMSc; Christian Torp-Pedersen, MD, DMSc

Background—The majority of out-of-hospital cardiac arrests (OHCAs) occur in residential locations, but knowledge about strategic placement of automated external defibrillators in residential areas is lacking. We examined whether residential OHCA areas suitable for placement of automated external defibrillators could be identified on the basis of demographic characteristics and characterized individuals with OHCA in residential locations.

Methods and Results—We studied 4828 OHCAs in Copenhagen between 1994 and 2005. The incidence and characteristics of OHCA were examined in every 100×100-m (109.4×109.4-yd) residential area according to its underlying demographic characteristics. By combining ≥2 demographic characteristics, it was possible to identify 100×100-m (109.4×109.4-yd) areas with at least 1 arrest every 5.6 years (characterized by >300 persons per area and lowest income) to 1 arrest every 4.3 years (characterized by >300 persons per area, lowest income, low education, and highest age). These areas covered 9.0% and 0.8% of all residential OHCAs, respectively. Individuals with OHCA in residential locations differed from public ones in that the patients were older (70.6 versus 60.6 years; $P<0.0001$), the ambulance response interval was longer (6.0 versus 5.0 minutes; $P<0.0001$), arrests occurred more often at night (21.2% versus 11.2%; $P<0.0001$), the patients had ventricular fibrillation less often (12.8% versus 38.1%; $P<0.0001$), and the patients had a worse 30-day survival rate (3.2% versus 13.9%; $P<0.0001$).

Conclusions—On the basis of simple demographic characteristics of a city center, we could identify residential areas suitable for automated external defibrillator placement. Individuals with OHCA in residential locations were more likely to have characteristics associated with poor outcome compared with public arrests. (*Circulation*. 2010;122:623-630.)

Key Words: cardiopulmonary resuscitation ■ death, sudden ■ defibrillation, electric ■ epidemiology

Automated external defibrillators (AEDs) placed in high-density public areas can save lives,¹⁻⁴ but 65% to 80% of out-of-hospital cardiac arrests (OHCAs) take place in residential areas and remain not covered by publicly placed AEDs.^{5,6} A few attempts have been made to place AEDs in residential areas, but the results have been disappointing.^{3,7,8} In addition, successful resuscitation is reported in only 2% of residential OHCAs.^{6,9} Public-access defibrillation (PAD) could be of value in residential areas if AEDs are strategically placed where many arrests occur. However, knowledge is lacking about strategic placement of AEDs in residential areas and the potential impact on overall survival. Therefore, we studied a large series of OHCAs in the city of Copenhagen to examine whether high-risk areas of OHCA suitable for

AED placement could be identified on the basis of the demographic characteristics of the area. Second, because the effect of PAD in residential areas is unknown, we compared individual patient characteristics according to location of cardiac arrest (residential versus public location) to find any important differences that could affect the likelihood of survival.

Editorial see p 567
Clinical Perspective on p 630

Methods

Study Setting

Central Copenhagen covers 97 km² (37.4 sq miles) and has a population of ≈600 000 people. In Copenhagen, the emergency

Received November 18, 2009; accepted May 20, 2010.

From the Department of Cardiology, Copenhagen University Hospital, Gentofte, Denmark (F.F., G.H.G., P.W., M.L.H., E.L.F., S.S.A., C.T.-P.); Emergency Medicine and Emergency Medical Services, Capital Region of Denmark (F.K.L.); Mobile Emergency Care Unit of Copenhagen, Capital Region of Denmark (S.L.N.); National Institute of Public Health, Copenhagen, Denmark (S.R.); and Department of Cardiology, Copenhagen University Hospital, Rigshospitalet, Denmark (T.K.S., L.K.).

Correspondence to Fredrik Folke, MD, Department of Cardiology, Copenhagen University Hospital, Gentofte, Niels Andersens Vej 65, 2900 Hellerup, Denmark. E-mail FF@heart.dk

© 2010 American Heart Association, Inc.

Circulation is available at <http://circ.ahajournals.org>

DOI: 10.1161/CIRCULATIONAHA.109.924423

medical system is 2 tiered. Ambulances with paramedics are dispatched from 7 different locations 24 hours a day and 7 days a week. In all cases of suspected OHCA, a physician-staffed mobile emergency care unit (MECU) is also dispatched, and OHCA data are registered. Two MECUs are available during daytime and on weekends; the rest of the time, 1 MECU is available. The MECUs are based at the main fire station located in the city center.

Study Population

The MECUs recorded 5420 OHCA between January 1994 and December 2005. Because of incomplete electronic registration in 1999, no arrests from that year are included. We excluded 544 patients with OHCA resulting from suicide, poisoning, drowning, trauma, or exsanguination or because the patient was judged terminally ill by the on-location emergency physician. Another 48 patients were excluded because the location of OHCA was unknown or because the arrest occurred outside the defined study area. The Utstein definitions¹⁰ for patient data collection were followed. OHCA were categorized as cardiac arrests occurring in public locations, defined as all areas accessible to the general public, all outdoor locations, public transportation sites, schools, outpatient clinics, and commercial and civic establishments, or as residential locations, defined as single- or multiple-family dwellings, senior housing, or nursing homes.

Individual patient characteristics were obtained from the MECUs and the National Patient Register, which contains information for hospitalizations since 1978.¹¹ Comorbidity was defined as previous hospital admissions up to 10 years before date of cardiac arrest, a method used previously.¹² Each admission is registered by 1 primary diagnosis and, if appropriate, ≥ 1 secondary diagnoses according to the *International Classification of Diseases* (10th revision). Information about patients' survival status (dead or alive) at least 30 days after the cardiac arrest was obtained from the Central Population Register.

Study Methods

We defined high-risk OHCA areas as those where at least 1 cardiac arrest occurred every 5 years within a 100×100-m (109.4×109.4-yd) area. The incidence rate chosen is in accordance with guidelines from the American Heart Association,¹³ and the size of these areas was regarded the maximum distance in which an AED placed in a residential area could be transported within 1½ minutes by alerted personnel walking at a brisk pace.¹³ The European Grid System, a defined and harmonized grid net for Europe with standardized location and size of grid cells,¹⁴ was used to count all cardiac arrests in every 100×100-m (109.4×109.4-yd) grid cell included in the study area.

Demographic Characteristics of the Cardiac Arrest Areas

To analyze OHCA occurrence in residential locations according to the demographic characteristics of the grid cells, information about age, level of education, population density, and household income (comprising all incomes subject to taxation) in every grid cell was obtained through Statistics Denmark. Information about education level was retrieved from the Integrated Student Register of Statistics Denmark. Lower education level was defined as education <10 years (corresponding to primary and lower secondary school), a definition used previously.¹⁵

In addition, the predominant type of residence for every inhabited grid cell was identified and categorized into 1 of 4 groups: residential neighborhoods (single-dwelling residences), terrace housings (including multiple-dwelling residences), apartment blocks, residence halls. The exact location of every OHCA was geographically coded with a geographic information system¹⁶ and marked on a digital city map. The number of arrests for every grid cell was summarized and then analyzed in accordance with the demographic characteristics of the given grid cell.

Table 1. Demographic Characteristics of Individuals With OHCA in Residential Versus Public Locations, 1994 to 2005

Demographics	Residential Locations (n=3554)	Public Locations (n=1274)	P
Mean age, year (SD)	70.6 (15.9)	60.6 (17.5)	<0.0001
Men, y (SD) (%)	68.4 (15.6)	58.5 (16.5)	<0.0001
Women, y (SD) (%)	73.3 (15.9)	67.3 (18.8)	<0.0001
Male sex, n (%)	2008 (56.5)	977 (76.7)	<0.0001
Age group, n (%)			
<50 y	400 (11.3)	347 (27.2)	<0.0001
50–70 y	1053 (29.6)	516 (40.5)	<0.0001
>70 y	2101 (59.1)	411 (32.3)	<0.0001
Time of cardiac arrest, n (%)			
7 AM–3 PM	1500 (42.2)	630 (49.5)	<0.0001
3–11 PM	1302 (36.6)	501 (39.3)	0.10
11 PM–7 AM	752 (21.2)	143 (11.2)	<0.0001
Interval between call to MECU and rhythm assessment, median (SD), min	6.0 (2.60)	5.0 (2.30)	<0.0001
Initial heart rhythm, n (%)			
VF or pVT	455 (12.8)	485 (38.1)	<0.0001
Asystole	1783 (50.2)	502 (39.4)	<0.0001
Pulseless electrical activity	467 (13.1)	145 (11.4)	0.11
Other/unknown rhythm*	847 (23.8)	142 (11.1)	<0.0001
Comorbidity,† n (%)			
Prior myocardial infarction	517 (14.6)	258 (20.3)	<0.0001
Ischemic heart disease‡	661 (18.6)	235 (18.5)	0.90
Heart failure	706 (19.9)	194 (15.2)	<0.005
Diabetes mellitus	372 (10.5)	71 (5.6)	<0.0001
COPD	527 (14.8)	60 (4.7)	<0.0001
Stroke	465 (13.1)	92 (7.2)	<0.0001
Cancer	366 (10.3)	61 (4.8)	<0.0001
Survival after 30 d, n (%)	115 (3.2)	177 (13.9)	<0.0001

pVT indicates pulseless ventricular tachycardia; COPD, chronic obstructive pulmonary disease.

*Other/unknown rhythm includes pace rhythms, bradyarrhythmia, and unknown rhythm.

†Defined as previous hospital admissions up to 10 years before date of cardiac arrest.

‡Excluding prior myocardial infarction.

Cost-Effectiveness Calculations

The yearly cost of an AED, including maintenance costs, cost of hospitalization, and valuations of the quality of life of cardiac arrest survivors, was identical to that we have described in detail previously.¹² All costs were adjusted to US dollars 2009 (denoted as \$) through the use of an inflation rate of 3%/y. We assumed that for each AED deployed, 4 individuals expected to be in the proximity of the AED location would each require 2 hours of cardiopulmonary resuscitation (CPR) and AED training per year with an aggregate training cost of \$330, in agreement with the figure used in a previous cost-effectiveness analysis by Cram et al.¹⁷ In Copenhagen, the rate of ventricular fibrillation (VF) was 12.8% among residential OHCA, with a median MECU response time of 6 minutes (Table 1). Assuming that AED deployment in residential areas would shorten AED response intervals by at least 1 minute, we estimated that the rate of VF would increase $\approx 10\%$, in accordance with findings of Herlitz et al^{9,18} and the figure used in other cost-effectiveness

studies.¹⁹ The life expectancy of cardiac arrest survivors >60 years of age has been reported to be ≈ 4 years in Copenhagen.²⁰ To estimate the cost per quality-adjusted life-year (QALY), we divided the total cost by the total estimated life-years gained adjusted for valuations of the quality of life. Because the survival effect with residential PAD is unknown, the calculations were based on arbitrarily chosen survival rates among patients found in VF of 15%, 25%, and 35%. This method is in accordance with that used in other cost-effectiveness studies.^{17,19}

Statistical Analysis

We used multivariable logistic regression to examine the likelihood of the patient having VF adjusted for sex, age, response time, comorbidity, time of day of OHCA, and location of OHCA (public location was used as the reference). Model assumptions (linearity of continuous variables and lack of interactions) were tested and found to be valid unless otherwise indicated. To estimate the rate ratio of OHCA for grid cells according to the underlying demographic characteristic, we used a modified Poisson model with a robust error variance.²¹ The cumulative number of OHCA in the grid cells served as the dependent variable, and the demographic characteristics of the grid cell were the independent variables. The estimated rate ratio for the demographic characteristic was calculated by taking the exponential of the regression coefficient in the Poisson regression because of the constant follow-up time for all grid cells (11 years). Multivariable Poisson regression models adjusted for the presence of all demographic characteristics were performed. All statistical calculations were performed with the SAS statistical software package, version 9.1.4 (SAS Institute Inc, Cary, NC).

Ethics

Compliance with Danish law on data security was maintained, and the study was approved by the Danish Data Protection Agency (No. 2008-41-2685). No ethical approval is required for retrospective register studies in Denmark.

Results

During the study period, 4828 OHCA were included; 3554 (74%) occurred in residential locations and 1274 (26%) in public locations. Baseline characteristics of individuals experiencing OHCA in residential and public locations are compared in Table 1. After adjustment for age, sex, response time, comorbidities, and time of day of OHCA in the multivariate logistic regression analysis, the likelihood of the patient having a shockable heart rhythm (VF/pulseless ventricular tachycardia) when the MECU arrived was markedly lower for arrests in residential compared with public locations (odds ratio [OR], 0.32; 95% confidence interval [CI], 0.27 to 0.38). However, there was a significant interaction between age and OHCA location ($P < 0.005$). In residential OHCA, the odds for shockable heart rhythm was lowest among patients >70 years of age (OR, 0.69; 95% CI, 0.49 to 0.97), whereas there was no significant difference for patients 50 to 70 years of age (OR, 1.36; 95% CI, 0.96 to 1.92) (reference age group, <50 years of age). For public OHCA, however, patients 50 to 70 years of age had the highest odds of a shockable heart rhythm (OR, 2.02; 95% CI, 1.47 to 2.79) but no significant difference for patients >70 years of age (OR, 1.21; 95% CI, 0.84 to 1.72). There were few dissimilarities relative to cardiac comorbidity, whereas chronic diseases like diabetes mellitus, chronic obstructive pulmonary disease, stroke, and cancer were significantly more pronounced among patients experiencing OHCA in residential locations compared with public ones, despite adjustment for age and sex (data not shown).

Figure 1 illustrates the unadjusted incidence rate of OHCA in residential locations by population density, income, age, and proportion of the population with lower education levels. Figure 1A shows a nearly linear association between population density and the number of OHCA per grid cell up to a density of 400 persons per grid cell (40 000 persons/km² or 103 600 persons/sq mile), after which the number of OHCA decreases despite increasing population density. In the multivariate-adjusted Poisson analyses of OHCA occurrence according to the demographic characteristics of the area (Table 2), population density had the strongest association with OHCA occurrence. Interactions from the 4 explanatory variables entered into the Poisson model were tested separately. We found that household income was modified by low education level ($P < 0.001$). Overall, the OHCA occurrence increased with decreasing household income; however, this association was more pronounced among the lowest educational levels.

Figure 2 shows the distribution of residential OHCA by type of residence. The rate of OHCA was not solely associated with increasing population density illustrated by a low cardiac arrest rate in areas dominated by residence halls (0.35 cardiac arrest per grid cell) despite their high population density (169 persons per grid cell, corresponding to 16 900 persons/km² or 43 770 persons/sq mile). This is also reflected by the incidence of cardiac arrest by type of residence (Table 3).

Table 4 shows implications of different AED deployment strategies for OHCA in residential areas, including estimated costs of adding PAD programs to the established emergency medical system. The average OHCA rate in residential areas was 1 arrest every 16.8 years. No single demographic characteristic yielded an arrest rate as frequent as that recommended by the American Heart Association for AED placement (≥ 1 arrest every 5 years); however, when ≥ 2 characteristics were combined, it was possible to identify areas with 1 arrest every 4.3 to 5.6 years. The corresponding costs per QALY differed from \$50 300 to \$96 100 according to type of AED strategy and percentage of estimated survival among patients in VF.

Discussion

According to the recommendations from the American Heart Association,¹³ AEDs should be placed where there is a high likelihood of sudden cardiac arrest (≥ 1 arrest every 5 years). Using simple demographic characteristics of a city center, we were able to identify residential high-risk areas of OHCA potentially suitable for AED deployment. Importantly, no single demographic characteristic (population density, average age, average income, or proportion of those with low education) could identify areas with a sufficiently high arrest incidence to merit AED deployment (Table 4). However, when we focused on areas characterized by a high population density combined with the lowest income quartile, we found an average OHCA incidence of 1 arrest every 5.6 years. These areas constituted 3.0% of all residential areas and covered 9.0% ($n=319$) of all OHCA in residential locations. Assuming that placement of 1 AED in every grid cell is sufficient, this strategy would require deployment of 163

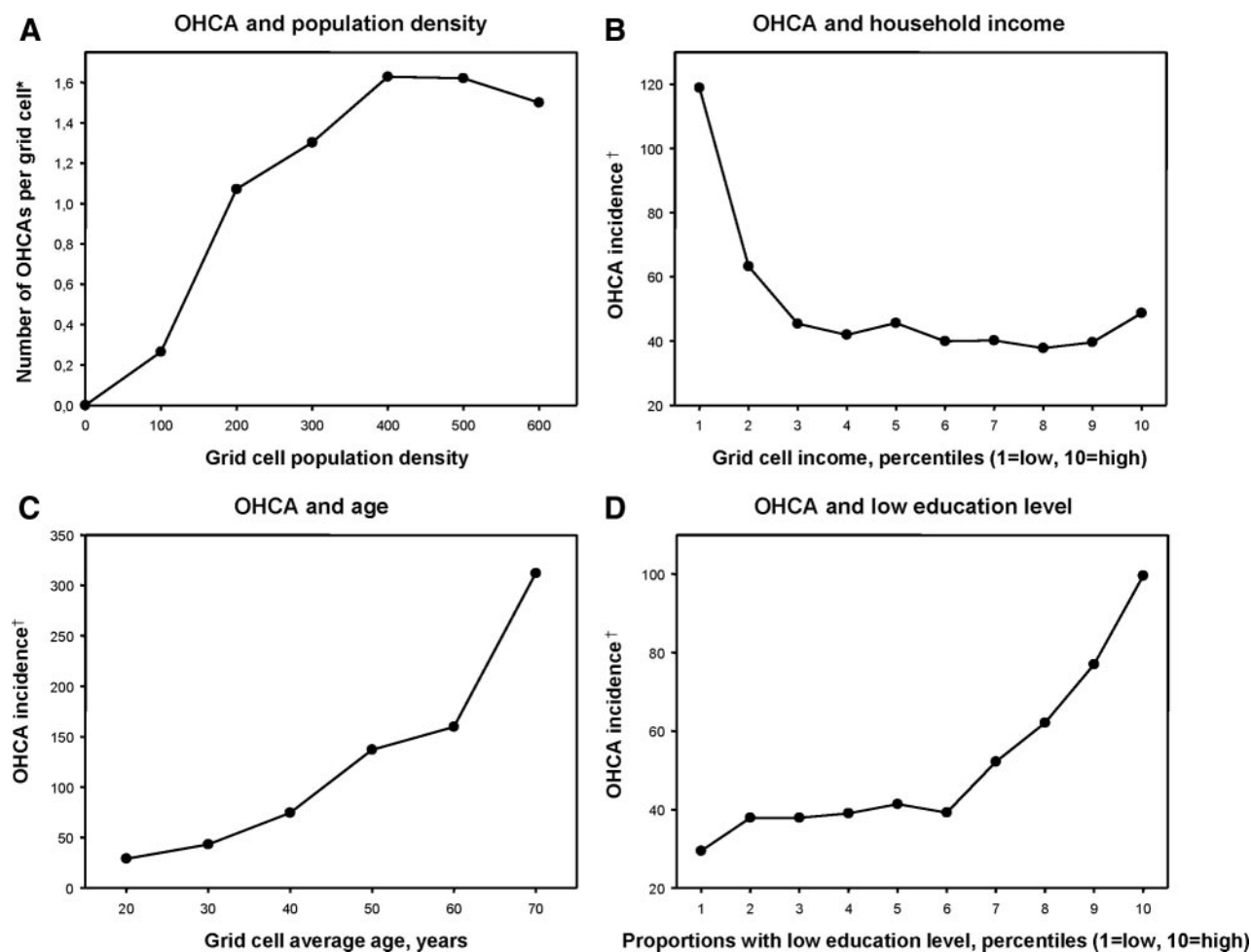


Figure 1. Incidence of OHCA according to population density (A), household income (B), average age (C), and proportion of the population with lower education levels (D). *Average number of OHCA per grid cell during 1994 to 2005. †OHCA incidence is defined as the number of arrests per 100 000 persons per year. Grid cells=100×100-m (109.4×109.4-yd) areas with uniform and standardized geographic locations.

AEDs. The arrest incidence reached 1 arrest every 4.9 years when the areas were additionally characterized by belonging to the highest quartile of low education. These areas covered 182 OHCA, 5.1% of all residential OHCA, and required placement of 81 AEDs. When the highest age quartile was also included, the arrest incidence reached 1 arrest every 4.3 years but covered only 28 OHCA.

The concept of strategic PAD deployment in residential areas has been addressed previously,²² but few studies have focused primarily on OHCA in residential locations. In the Home Automated External Defibrillator Trial (HAT), 7001 patients with previous anterior wall myocardial infarctions who were not candidates for an implantable cardioverter-defibrillator were randomly assigned to receive either standard care or standard care plus a home AED.⁷ Disappointingly, access to a home AED did not significantly improve overall survival, which was partly explained by the very low event rate of sudden cardiac arrest. Whereas the home AED strategy provides the opportunity to treat only those in the immediate household, placing AEDs outside people's homes provides the opportunity to use an AED in treating any one of many persons at risk. Because the rate of cardiac arrest was

<1%/y in HAT (corresponding to each AED having a possibility to be used in 1 arrest every 100 years), the chance of AED use would not exceed 1%/y despite the fact that an AED was used in every event. The chance of AED use could be as high as 20%/y if AEDs were placed in selected residential areas characterized by having at least 1 arrest every 4.3 to 5.6 years (Table 4). In a recent study of nationwide PAD in Japan,²³ a significantly shorter time from collapse to first defibrillation or CPR was observed for PAD compared with the emergency medical system. In addition, that study documented that if the number of PADs increased from 1 to >4/km², the rate of survival with minimal neurological impairment increased ≈4 times. These data support the AHA recommendations of AED availability within a 1½-minute walking distance,¹³ the same concept used in our study with AED deployment in every 100×100-m residential area.

Although several studies have found that AED deployment is cost-effective when placed in selected public locations,^{12,17,24,25} little is known about how to identify residential areas with the highest risk of OHCA, let alone the possible costs of PAD implementation in residential areas. The esti-

Table 2. Multivariate Adjusted Rate Ratios of OHCA in Residential Areas in Relation to the Demographic Characteristics of the Area

Grid Cell Measure	Rate Ratio of OHCA	95% CI
Average age		
Quartile 1 (younger)	1.00*	1.00*
Quartile 2	1.23	1.08–1.38
Quartile 3	1.23	1.09–1.39
Quartile 4 (older)	1.80	1.63–1.99
Household income		
Quartile 1 (low)	1.48	1.20–1.82
Quartile 2	1.03	0.84–1.27
Quartile 3	0.98	0.81–1.19
Quartile 4 (high)	1.00*	1.00*
Proportion with lower education level		
Quartile 1 (low)	1.00*	1.00*
Quartile 2	1.22	1.03–1.45
Quartile 3	1.37	1.16–1.63
Quartile 4 (high)	1.94	1.63–2.31
Population density		
Quartile 1 (low)	1.00*	1.00*
Quartile 2	2.27	1.82–2.83
Quartile 3	4.74	3.84–5.84
Quartile 4 (high)	8.61	7.00–10.58

Grid cells are defined as 100×100-m (109.4×109.4-yd) areas. All rate ratios are adjusted for average age, household income, proportion of low education, and population density of the grid cells.

*Reference groups.

mated costs for a future interventional trial presented in Table 4 clearly indicate that despite a focused strategy of AED placement in highly selected residential areas, the cost per QALY of such PAD implementation is likely to exceed the corresponding costs for PAD in selected public locations. Placing AEDs in areas with >300 persons per grid cell ($\approx 30\,000$ persons/km²) in combination with the lowest income quartile yielded cost per QALY ranging from \$55 000 to \$96 000 according to estimated VF survival. These areas constituted 9.0% (n=319) of all residential OHCA. In addition, Table 4 demonstrates that the estimated cost per QALY would be excessive if AED deployment were done at random. Likewise, although areas dominated by residence halls had the highest population density (Figure 2), the cost per QALY would greatly exceed \$100 000 for PAD in such areas. It is important to note, however, that the presented costs are rough estimates of the expenses combined with an interventional trial to test residential PAD. In addition, the costs should be used cautiously because they are sensitive to changes in several parameters, including the rate of VF, the life expectancy of OHCA survivors, and the failure to use an available AED on a nearby OHCA individual. Furthermore, there is ongoing debate about how much society should be willing to pay for a health intervention, but an amount of \$50 000 to \$100 000 per QALY gained is often cited in the literature as the cost-effectiveness threshold.^{25,26}

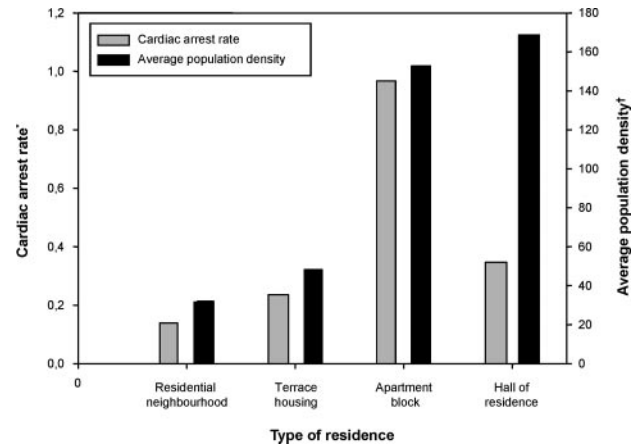


Figure 2. Incidence of OHCA in residential locations by type of residence and population density. *Defined as number of residential OHCA per 100×100-m (109.4×109.4-yd) grid cell through 1994 to 2005. †Defined as the average number of persons per 100×100-m (109.4×109.4-yd) grid cell.

Although PADs in residential areas could reduce time to defibrillation and thereby increase the chance of survival, differences in patient characteristics might influence the success compared with OHCA in public locations. In our study, individuals who experienced OHCA in residential locations were more likely to have characteristics that predicted a poor outcome, including older age,^{9,27} male sex,²⁸ arrests that occurred at night, a longer MECU response time,⁹ and less likelihood of a shockable heart rhythm compared with OHCA in public locations.^{9,27} The important differences according to location of arrest suggest that residential locations are more likely to have an adverse outcome, which is consistent with our observed OHCA survival rates of 13.9% and 3.2% for public and residential locations, respectively. A study by Herlitz et al⁹ suggested possible differences in comorbidity to be responsible for the adverse outcome among residential OHCA. Notably, we found that comorbidity among OHCA patients in residential versus public locations was consistent with cardiac diseases, whereas patients who experienced arrests in residential locations had concomitant chronic diseases more frequently (Table 1). Even when these differences were taken into account in the multivariate analysis, we found that the chance of having a shockable heart rhythm in residential locations was one third that of public ones. This finding could also be due to a longer interval from collapse to call for an ambulance in residential areas. Such data were not available in our study; however, a previous survey has estimated this interval to be at least 1 minute longer for residential compared with public locations.⁹

In the multivariate regression analysis, grid cells featuring a high level of crowding had the strongest association with OHCA occurrence (Table 2), but average age and income and the percentage with lower education levels were also associated with OHCA occurrence. Nevertheless, placing AEDs exclusively in areas with high population density reached an arrest incidence of only 1 arrest every 6.8 years. This could be explained by the fact that the majority of residence halls, inhabited by young people who are unlikely to experience OHCA (Figure 2 and Table 3), belonged to these crowded

Table 3. Characteristic of Residential Areas in Copenhagen in Relation to Type of Residence and OHCA Occurrence

Characteristics	Residential Neighborhoods	Terrace Housing	Apartment Blocks	Residence Halls
OHCA incidence*	41.2	43.2	56.6	18.6
Proportion of all habitations, %	27.3	9.1	62.6	1.0
Proportion of the total study area, %	14.0	4.7	32.5	0.5
Average population density, persons/grid cell	31.6	48.2	152.7	168.8
Average household income,† \$	101 000	106 000	58 000	49 000
Individuals with lower education level, %	10.4	11.9	20.0	12.7
Proportion of total population, %	7.8	4.0	86.7	1.5
Proportion of all residential OHCA, %	5.9	3.1	90.4	0.5

*Defined as the number of residential OHCA per year per 100 000 persons.

†Household income is in 2005 US dollars rounded to nearest thousand.

areas. In contrast with our findings, an American study found that regional median income was the predominant factor associated with cardiac arrest rate.²⁹ This was partly explained by lower-income areas having reduced access to health care. Such a disparity was minimized in our study because the healthcare system in Denmark covers all medical services provided by hospitals and physicians with no user fees or copayment. Nevertheless, our results showing increased OHCA rates in disadvantaged neighborhoods seem plausible because previous studies have reported that the socioeconomic environment of neighborhoods and health-

related behaviors such as smoking, obesity, and physical activity are associated with lower income levels³⁰; likewise, an increased incidence of coronary heart disease is also observed.³¹

If PAD is to succeed in residential areas, several obstacles in addition to strategic AED deployment need to be resolved. It is unrealistic to assume that an AED is immediately present on every street corner or in every staircase. Furthermore, strategic placement of AEDs alone is insufficient for successful PAD but should be part of a more comprehensive emergency care plan,¹³ which includes training potential

Table 4. Implications of Different AED Deployment Strategies for OHCA in Residential Areas

AED Deployment Principle	OHCAs, n (% of All OHCA)	AEDs Needed, n	Years for ≥1 OHCAs	Estimated Cost per QALY* if VF Survival Is		
				15%	25%	35%
In all inhabited areas	3554 (100)	5390	16.8	208 700	115 600	87 800
By population density, n persons/grid cell†						
<100	853 (24.0)	3213	41.4	460 600	231 300	162 900
>100	2701 (76.0)	2177	8.9	129 100	79 000	64 100
>200	1463 (41.1)	1031	7.8	117 800	73 800	60 700
>300	601 (16.9)	370	6.8	107 800	69 200	57 700
>400	146 (4.1)	91	6.9	108 00	69 500	58 000
By type of residence						
Residential neighborhoods	211 (5.9)	1471	76.7	819 400	396 200	269 900
Terrace housing	112 (3.1)	489	48.0	527 700	262 000	183 000
Apartment blocks	3213 (90.4)	3378	11.6	156 600	91 700	72 300
Residence halls	18 (0.5)	52	31.8	362 300	186 200	133 600
In areas with the lowest income quartile	1839 (51.7)	1347	8.1	120 900	75 300	61 600
In areas with the highest age quartile	1366 (38.4)	1215	9.8	138 500	83 300	66 900
In areas with the highest quartile of low education	1627 (45.8)	1294	8.7	128 000	78 500	63 700
In areas with >300 persons/grid cell and lowest income	319 (9.0)	163	5.6	96 100	63 900	54 200
In areas with >300 persons/grid cell, lowest income, and lower education	182 (5.1)	81	4.9	88 700	60 500	52 000
In areas with >300 persons/grid cell, lowest income, lower education, and highest age	28 (0.8)	11	4.3	82 900	57 800	50 300

*All costs are 2009 US dollars rounded to nearest hundred.

†Grid cells are defined as 100×100-m (109.4×109.4-yd) areas with uniform and standardized geographic locations.

rescuers in CPR and AED use. Bystander CPR can triple survival rates,³² and familiarity with AED use, along with an increased AED dissemination, might promote both a willingness to use an AED and a positive attitude toward CPR.³³ In addition, establishing a local AED registry with information on location and availability of AEDs in the community would allow the emergency dispatcher to direct a rescuer to the nearest AED. In selected areas suitable for AED deployment, AEDs could be placed strategically in the middle of several blocks or streets or near high-rise apartments. Potential rescuers in these areas could include security personnel or designated “cardiac arrest caretakers” trained in CPR and defibrillation²² but also with the possibility of AED access by untrained laypersons. Whenever a cardiac arrest call comes into the emergency dispatch center, it could immediately lead to dispatch of both the MECU and the trained laypersons at the given location. Such a setup could potentially reduce the failure to use an AED on a nearby individual with OHCA, which occurred in 65% of all cases in the Public Access Defibrillation trial.³

This study has several limitations. First, because this was not a prospective, randomized study, we cannot determine whether strategic placement of AEDs in residential areas along with a detailed emergency plan would actually increase the survival of OHCA. This remains to be evaluated as part of a randomized, interventional trial of residential PADs. Such a trial would also allow an accurate assessment of the cost-effectiveness associated with residential PAD. Second, the geographic unit used in this study does not account for complex and varied local logistics that would accompany actual AED placement within a community. Therefore, our use of grid cells might not be representative of demographic characteristics compared with other communities with different demographics and infrastructure. Third, owing to incomplete data on the rate of witnessed arrest and bystander CPR, we could not examine how these factors affected the observed rate of VF and 30-day survival in residential and public OHCA locations.

Conclusions

Using simple demographic characteristics of a city center, we were able to identify residential high-risk areas of OHCA suitable for AED placement. Such areas comprised <3% of all residential quarters but included up to 9% of all residential OHCA. However, individuals with OHCA in residential locations were more likely to have characteristics associated with a poor outcome, a finding emphasizing the necessity for strategic placement of AEDs if future PAD programs in residential areas are to succeed without excessive costs.

Source of Funding

This research was funded by an unrestricted grant from the Tryg Foundation (Denmark). The sponsor had no role in the design and conduct of the study; in the collection, management, analysis, and interpretation of the data; or in the preparation, review, or approval of the manuscript.

Disclosures

None.

References

- Caffrey SL, Willoughby PJ, Pepe PE, Becker LB. Public use of automated external defibrillators. *N Engl J Med*. 2002;347:1242–1247.
- Valenzuela TD, Roe DJ, Nichol G, Clark LL, Spaite DW, Hardman RG. Outcomes of rapid defibrillation by security officers after cardiac arrest in casinos. *N Engl J Med*. 2000;343:1206–1209.
- Hallstrom AP, Ornato JP, Weisfeldt M, Travers A, Christenson J, McBurnie MA, Zalenski R, Becker LB, Schron EB, Proschan M. Public-access defibrillation and survival after out-of-hospital cardiac arrest. *N Engl J Med*. 2004;351:637–646.
- Davies CS, Colquhoun MC, Boyle R, Chamberlain DA. A national programme for on-site defibrillation by lay people in selected high risk areas: initial results. *Heart*. 2005;91:1299–1302.
- Eisenberg MS, Mengert TJ. Cardiac resuscitation. *N Engl J Med*. 2001;344:1304–1313.
- Norris RM. Circumstances of out of hospital cardiac arrest in patients with ischaemic heart disease. *Heart*. 2005;91:1537–1540.
- Bardy GH, Lee KL, Mark DB, Poole JE, Toff WD, Tonkin AM, Smith W, Dorian P, Packer DL, White RD, Longstreth WT Jr, Anderson J, Johnson G, Bischoff E, Yallop JJ, McNulty S, Ray LD, Clapp-Channing NE, Rosenberg Y, Schron EB. Home use of automated external defibrillators for sudden cardiac arrest. *N Engl J Med*. 2008;358:1793–1804.
- Eisenberg MS, Moore J, Cummins RO, Andersen E, Litwin PE, Hallstrom AP, Hearne T. Use of the automatic external defibrillator in homes of survivors of out-of-hospital ventricular fibrillation. *Am J Cardiol*. 1989;63:443–446.
- Herlitz J, Eek M, Holmberg M, Engdahl J, Holmberg S. Characteristics and outcome among patients having out of hospital cardiac arrest at home compared with elsewhere. *Heart*. 2002;88:579–582.
- Cummins RO, Chamberlain DA, Abramson NS, Allen M, Baskett PJ, Becker L, Bossaert L, Deloos HH, Dick WF, Eisenberg MS, Evans TR, Holmberg S, Kerber R, Mullie A, Ornato JP, Sandoe E, Skulberg A, Tunstall-Pedoe H, Swanson R, Thies WH. Recommended guidelines for uniform reporting of data from out-of-hospital cardiac arrest: the Utstein Style: a statement for health professionals from a task force of the American Heart Association, the European Resuscitation Council, the Heart and Stroke Foundation of Canada, and the Australian Resuscitation Council. *Circulation*. 1991;84:960–975.
- Andersen TF, Madsen M, Jorgensen J, Mellemejkær L, Olsen JH. The Danish National Hospital Register: a valuable source of data for modern health sciences. *Dan Med Bull*. 1999;46:263–268.
- Folke F, Lippert FK, Nielsen SL, Gislason GH, Hansen ML, Schramm TK, Sorensen R, Fosbol EL, Andersen SS, Rasmussen S, Kober L, Torp-Pedersen C. Location of cardiac arrest in a city center: strategic placement of automated external defibrillators in public locations. *Circulation*. 2009;120:510–517.
- Aufderheide T, Hazinski MF, Nichol G, Steffens SS, Buroker A, McCune R, Stapleton E, Nadkarni V, Potts J, Ramirez RR, Eigel B, Epstein A, Sayre M, Halperin H, Cummins RO. Community lay rescuer automated external defibrillation programs: key state legislative components and implementation strategies: a summary of a decade of experience for healthcare providers, policymakers, legislators, employers, and community leaders from the American Heart Association Emergency Cardiovascular Care Committee, Council on Clinical Cardiology, and Office of State Advocacy. *Circulation*. 2006;113:1260–1270.
- Annoni A. European reference grids: proposal for a European grid system. In: Workshop Proceedings and Recommendations for the Institute for Environment and Sustainability; 2005. EUR Report 21494 EN.
- Rasmussen JN, Gislason GH, Rasmussen S, Abildstrom SZ, Schramm TK, Kober L, Diderichsen F, Osler M, Torp-Pedersen C, Madsen M. Use of statins and beta-blockers after acute myocardial infarction according to income and education. *J Epidemiol Community Health*. 2007;61:1091–1097.
- Experian Denmark. NeoZone: X-Point Office. Available at: <http://www.neozone.com/web/default.asp>. Accessed April 20, 2009.
- Cram P, Vijan S, Fendrick AM. Cost-effectiveness of automated external defibrillator deployment in selected public locations. *J Gen Intern Med*. 2003;18:745–754.
- Herlitz J, Engdahl J, Svensson L, Young M, Angquist KA, Holmberg S. Decrease in the occurrence of ventricular fibrillation as the initially observed arrhythmia after out-of-hospital cardiac arrest during 11 years in Sweden. *Resuscitation*. 2004;60:283–290.
- Foutz RA, Sayre MR. Automated external defibrillators in long-term care facilities are cost-effective. *Prehosp Emerg Care*. 2000;4:314–317.

20. Holler NG, Mantoni T, Nielsen SL, Lippert F, Rasmussen LS. Long-term survival after out-of-hospital cardiac arrest. *Resuscitation*. 2007;75:23–28.
21. McCullagh PNJ. *Generalized Linear Models, Second Edition (Monographs on Statistics and Applied Probability)*. London, UK: Chapman & Hall; 1989.
22. Zipes DP. President's page: the Neighborhood Health Watch Program: Save A Victim Everywhere (SAVE). *J Am Coll Cardiol*. 2001;37:2004–2005.
23. Kitamura T, Iwami T, Kawamura T, Nagao K, Tanaka H, Hiraide A. Nationwide public-access defibrillation in Japan. *N Engl J Med*. 362:994–1004.
24. Nichol G, Valenzuela T, Roe D, Clark L, Huszti E, Wells GA. Cost effectiveness of defibrillation by targeted responders in public settings. *Circulation*. 2003;108:697–703.
25. Nichol G, Huszti E, Birnbaum A, Mahoney B, Weisfeldt M, Travers A, Christenson J, Kuntz K. Cost-effectiveness of lay responder defibrillation for out-of-hospital cardiac arrest. *Ann Emerg Med*. 2009;54:226–235, e221–e222.
26. Laupacis A, Feeny D, Detsky AS, Tugwell PX. How attractive does a new technology have to be to warrant adoption and utilization? Tentative guidelines for using clinical and economic evaluations. *CMAJ*. 1992;146:473–481.
27. Swor RA, Jackson RE, Compton S, Domeier R, Zalenski R, Honeycutt L, Kuhn GJ, Frederiksen S, Pascual RG. Cardiac arrest in private locations: different strategies are needed to improve outcome. *Resuscitation*. 2003;58:171–176.
28. Herlitz J, Svensson L, Engdahl J, Gelberg J, Silfverstolpe J, Wisten A, Angquist KA, Holmberg S. Characteristics of cardiac arrest and resuscitation by age group: an analysis from the Swedish Cardiac Arrest Registry. *Am J Emerg Med*. 2007;25:1025–1031.
29. Feero S, Hedges JR, Stevens P. Demographics of cardiac arrest: association with residence in a low-income area. *Acad Emerg Med*. 1995;2:11–16.
30. Diez-Roux AV, Nieto FJ, Muntaner C, Tyroler HA, Comstock GW, Shahar E, Cooper LS, Watson RL, Szklo M. Neighborhood environments and coronary heart disease: a multilevel analysis. *Am J Epidemiol*. 1997;146:48–63.
31. Diez Roux AV, Merkin SS, Arnett D, Chambless L, Massing M, Nieto FJ, Sorlie P, Szklo M, Tyroler HA, Watson RL. Neighborhood of residence and incidence of coronary heart disease. *N Engl J Med*. 2001;345:99–106.
32. Holmberg M, Holmberg S, Herlitz J. Effect of bystander cardiopulmonary resuscitation in out-of-hospital cardiac arrest patients in Sweden. *Resuscitation*. 2000;47:59–70.
33. Kuramoto N, Morimoto T, Kubota Y, Maeda Y, Seki S, Takada K, Hiraide A. Public perception of and willingness to perform bystander CPR in Japan. *Resuscitation*. 2008;79:475–481.

CLINICAL PERSPECTIVE

Whereas a tremendous amount of resources has focused on deployment of automated external defibrillators (AEDs) in public locations, the majority of out-of-hospital cardiac arrests (OHCAs) take place in residential areas and remain not covered by publicly placed AEDs. Furthermore, little is known about how to identify residential areas with the highest risk of OHCA, let alone the possible cost of public-access defibrillation in such areas. We therefore performed a systematic analysis of OHCAs in residential areas of Copenhagen to examine whether high-risk areas of OHCA suitable for AED placement could be identified. Using simple demographic characteristics of a city center, we were able to identify residential areas characterized by having at least 1 OHCA every 4.3 to 5.6 years within a 100×100-m area. These areas comprised <3% of all residential quarters but included up to 9% of all residential arrests. The estimated cost of AED deployment in selected residential areas was below \$100 000 per quality-adjusted life-year but likely to exceed the corresponding cost for AED deployment in public locations. Furthermore, individuals with OHCAs in residential locations were more likely to have characteristics associated with a poor outcome compared with public ones in that the patients were older and more often male, the ambulance response interval was longer, arrests occurred more often at night, and the patients had ventricular fibrillation less often. Therefore, if future public-access defibrillation programs in residential areas are to succeed without excessive costs, strategic placement of AEDs in selected residential areas is a necessity.